STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 8- Poisson Approximation to Binomial Distribution

Dr. Gabriel Kallah-Dagadu

Department of Statistics and Actuarial Science
University of Ghana

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Session Outline

1. Poisson Approximation to Binomial

2. Applications

Session Overview

In the previous session, students were introduced to Poisson probability distribution. This session discussed Poisson approximation to the Binomial distribution. Students are introduced to real-life problems where calculating a probability of an event which follows a Binomial distribution can be computed using Poisson approximation.

Learning Outcomes

At the end of the session, students will be able to:

1. Derive Poisson approximation to the Binomial distribution.
2. Apply Poisson approximation to the Binomial to solve problems.

Relation Between the Binomial and Poisson

- In the binomial distribution, if n is large while the probability p of occurrence of an event is close to zero, so that $q = 1 - p$ is close to 1, the event is called a *rare event*.

- n practice we shall consider an event as rare if the number of trials is at least 50 ($n \geq 50$), while np is less than 5.

- For such cases the binomial distribution is very closely approximated by the Poisson distribution, with $\lambda = np$.

- The expected result is that, by placing $\lambda = np$, $q \approx 1$ and $p \approx 0$.

Poisson Approximation to Binomial

Theorem 1:

Suppose that the random variable $X \sim b(n,p)$ and the probability mass function given as:

$$P(X=x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0,1,2,\ldots,n;\ 0 < p < 1.$$

Suppose that n is large ($i.e.\ n \to \infty$) as p is small ($i.e.\ p \to 0$), then we let $\lambda = np$, a constant. Under the above conditions it implies that

$$\lim_{n\to\infty} P(X=x) = \lim_{n\to\infty} \binom{n}{x} p^x q^{n-x}$$

$$\approx \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0,1,2,\ldots$$

Proof:

Given that $X \sim b(n, p)$ we have:

$$P(X = x) = \begin{pmatrix} n \\ x \end{pmatrix} p^x\, q^{n-x}$$

$$= \frac{n!}{x!(n-x)!} p^x\, q^{n-x}$$

$$= \frac{n(n-1)(n-2)\ldots(n-x-1)(n-x)!}{x!(n-x)!} p^x\, q^{n-x}$$

$$= \frac{n(n-1)(n-2)\ldots(n-x-1)}{x!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

Note that $p = \frac{\lambda}{n}$, since $\lambda = np$.

$$= n^x \frac{1(1-\frac{1}{n})(1-\frac{2}{n})\ldots(1-\frac{x+1}{n})}{x!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

Proof Cont'd

$$P(X=x) = n^x \frac{1(1-\frac{1}{n})(1-\frac{2}{n})\dots(1-\frac{x+1}{n})}{x!}\frac{\lambda^x}{n^x}\left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{\lambda^x}{x!}\left[\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{x+1}{n}\right)\right]\left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{\lambda^x}{x!}\left[\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{x+1}{n}\right)\right]\left(1-\frac{\lambda}{n}\right)^{n}\left(1-\frac{\lambda}{n}\right)^{-x}$$

Now, taking the limits as $n \to \infty$

$$\lim_{n\to\infty} P(X=x) = \lim_{n\to\infty}\left\{\frac{\lambda^x}{x!}\left[\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{x+1}{n}\right)\right]\right\} \quad (1)$$

$$\times \lim_{n\to\infty}\left\{\left(1-\frac{\lambda}{n}\right)^{n}\left(1-\frac{\lambda}{n}\right)^{-x}\right\} \quad (2)$$

Proof Cont'd

$$\lim_{n\to\infty} P(X=x) = \frac{\lambda^x}{x!}\lim_{n\to\infty}\left[\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{x+1}{n}\right)\right]$$
$$\times \lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^n \times \lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^{-x}$$

However,

$$\lim_{n\to\infty}\left[\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{x+1}{n}\right)\right] = 1,$$

$$\lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^{-x} = 1 \quad \text{and}$$

$$\lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^{n} = e^{-\lambda}, \quad \text{from a well known theorem.}$$

$$\therefore \quad \lim_{n\to\infty} P(X=x) = \frac{\lambda^x}{x!}e^{-\lambda}, \quad \text{where } x = 0,1,2,3,\ldots$$

$\square$

Examples

Question 1

Suppose that at a busy traffic junction the probability p of an individual car having an accident is 0.0001. During a certain part of the day, 1000 cars pass through the junction. What is the probability of two or more cars being involved in an accident within this period?

Solution:

Let X represent the number of cars involved in an accident within a day at the traffic junction. Then the random variable $X \sim b(n = 1000, p = 0.0001)$. Therefore, the required probability is $P(X \geq 2)$, which is tedious to calculate using the binomial distribution. Therefore using the Poisson approximation to the Binomial,we have; $\lambda = np = 1000 \times 0.0001 = 0.1$.

Solution:

Hence the required probability is

$$P(X \geq 2) = 1 - P(X < 2) = 1 - \sum_{x=0}^{1} \frac{\lambda^x e^{-\lambda}}{x!}$$

$$= 1 - \sum_{x=0}^{1} \frac{0.1^x e^{-0.1}}{x!}$$

$$= 1 - (1 + 0.1)\, e^{-0.1} = 1 - 0.9953$$

$$= 0.00468$$

□

Question 1

Ten percent of the tools produced in a certain manufacturing process turn out to be defective. Find the probability that in a sample of 10 tools chosen at random, exactly 2 will be defective, by using

a. the binomial distribution,
b. the Poisson approximation to the binomial distribution.

Solution

(a)

The probability of a defective tool is $p = 0.1$. Let X denote the number of defective tools out of 10 chosen. Then, according to the binomial distribution,

$$P(X = 2) = \binom{10}{2}(0.1)^2(0.9)^8 = 0.1937 \approx 0.19$$

(b)

We have $\lambda = np = (10)(0.1) = 1$. Then, according to the Poisson distribution,

$$P(X = 2) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{1^2 e^{-1}}{2!} = 0.1839 \approx 0.18$$

In general, the approximation is good if $p \leq 0.1$ and $\lambda = np \leq 5$.

Question 2

If the probability that an individual will suffer a bad reaction from injection of a given serum is 0.001. Determine the probability that out of 2000 individuals,

1. (i) exactly 3,
2. (ii) more than 2,

individuals will suffer a bad reaction.

Solution

Let X denote the number of individuals suffering from a bad reaction. X is Bernoulli distributed, but since bad reactions are assumed to be rare events, we can suppose that X is Poisson distributed, i.e.,

$$P(X = x) = \frac{\lambda^x \, e^{-\lambda}}{x!}, \quad \text{where } \lambda = np = 2000(0.001) = 2.$$

1. The probability that exactly 3 will suffer bad reaction is

$$P(X = 3) = \frac{2^3 \, e^{-2}}{3!} = 0.180$$

2. The probability that more than 2 individuals will suffer bad reaction is

$$P(X > 2) = 1 - P(X \le 2) = 1 - [p(0) + p(1) + p(2)]$$

Solution

ii. The probability that more than 2 individuals will suffer bad reaction is

$$P(X > 2) = 1 - P(X \leq 2) = 1 - [p(0) + p(1) + p(2)]$$

$$= 1 - \left[\frac{2^0\ e^{-2}}{0!} + \frac{2^1\ e^{-2}}{1!} + \frac{2^2\ e^{-2}}{2!}\right]$$

$$= 1 - 5e^{-2} = 0.323$$

Trial Questions

1. If 5% of the electric bulbs manufactured by a company are defective, find the probability that
   a. more than 5,
   b. between 1 and 3,
   c. less than or equal to 2,

   bulbs will be defective.

2. A bag contains 1 red and 7 white marbles. A marble is drawn from the bag, and its color is observed. Then the marble is put back into the bag and the contents are thoroughly mixed. Using
   a. the binomial distribution,
   b. the Poisson approximation to the binomial distribution,

   find the probability that in 8 such drawings, a red ball is selected exactly 3 times.

3. According to the National office of the Vital Statistics of the Ghana Statistical Service, the average number of accidental drownings per year in Ghana is 5.0 per 100,000 population. Find the probability that in a city of population 200,000 there will be
    a. 0,
    b. 2,
    c. 6,
    d. 8,
    e. between 4 and 8,
    f. fewer than 3,

    accidental drownings per year.

Trial Questions

4. An insurance company has 40000 clients covered for “severe industrial accident”. Such accidents are estimated to affect 1 in 200000 of the population in any year, and they are assumed to occur independently of each other.
   Let X be the number of claims for severe industrial accident received by the company in any one year.

   a. State the distribution of X, and explain why an appropriate Poisson would give a good approximation.

   b. Find the probability that the number of claims received by the company in a year is at least 2.

End of Session

THANK YOU!!!